ModelIt!

Teacher's Guide

What Makes Wind Blow?

The Invisible Force That Moves the Air

Quick Reference

Grade Level	Grade 2
NGSS Standard	2-ESS2-3
Time	45–60 minutes (core lesson)
Materials	Student Activity Pack, devices with ModelIt! access
Prep	Print activity packs, test ModelIt! on student devices
Grouping	Partners or small groups (2–4)

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1 NGSS Standards & CAST Alignment

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Performance Expectation: 2-ESS2-3

Obtain information to identify where water is found on Earth and that it can be solid or liquid.

Three-Dimensional Learning

Science Practice	**Obtaining, Evaluating, and Communicating Information** — Students obtain information about wind patterns and communicate observations about how wind affects objects in the environment.
Disciplinary Core Idea	**ESS2.D Weather and Climate** — Weather is the combination of sunlight, wind, snow or rain, and temperature in a particular region at a particular time. Wind is one of the key components of weather.
Crosscutting Concept	**Cause and Effect** — Students identify that uneven heating by the sun causes air to move, creating wind as a direct effect.

CAST Testing Alignment

MULTIPLE CHOICE:

On a hot summer day at the beach, you feel a cool breeze blowing from the water toward the sand. Why does the breeze blow from the water to the sand?

CONSTRUCTED RESPONSE:

Explain what wind is and what causes it. Use the words 'sun,' 'warm air,' and 'cool air' in your answer.

2 Scoring Rubric

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Assessment Reference

This is the same rubric students see in their Activity Pack. Use it during and after the lesson for formative and summative assessment. Circle or highlight the proficiency level for each skill.

Skill	Expert (4)	Proficient (3)	Developing (2)	Beginning (1)
Identifying Components	EXPERT Correctly identifies and sorts ALL components as internal or external with detailed reasoning	PROFICIENT Correctly identifies and sorts MOST components with some reasoning	DEVELOPING Identifies components but has difficulty sorting or explaining	BEGINNING Needs significant help identifying components
Establishing Relationships	EXPERT Accurately describes ALL positive and negative relationships with clear evidence	PROFICIENT Accurately describes MOST relationships with some evidence	DEVELOPING Identifies some relationships but struggles with direction	BEGINNING Needs help understanding how components connect
Running Simulations	EXPERT Makes accurate predictions AND explains surprising results using evidence	PROFICIENT Makes reasonable predictions and records clear observations	DEVELOPING Runs simulations but predictions or observations are incomplete	BEGINNING Needs help running simulations or recording data
Explaining Results	EXPERT Uses multiple pieces of evidence to explain HOW and WHY systems interact	PROFICIENT Uses evidence to explain system interactions	DEVELOPING Gives partial explanations with limited evidence	BEGINNING Needs help connecting evidence to explanations

Scoring Guidance

Expert (4): Student demonstrates deep understanding and can transfer knowledge to new contexts.
Proficient (3): Student meets grade-level expectations and shows solid understanding.
Developing (2): Student is making progress but needs additional scaffolding or practice.
Beginning (1): Student needs significant support; consider using Differentiation strategies (Section 7).

3 Background Content Knowledge

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For Your Confidence

This section provides the science background you need to facilitate this lesson with authority. You don't need to teach all of this — it's here so you can answer student questions accurately.

Wind is one of the most common weather phenomena on Earth, yet most people never think about what actually causes it. Understanding wind helps us predict weather, design buildings, and even generate electricity.

What Is Wind?

Wind is simply air in motion. Air is real matter — it has weight and takes up space, even though we cannot see it. When air moves from one place to another, we feel it as wind. A gentle wind is called a breeze, and a very strong wind can be a gust or even a storm. Wind can be as soft as a whisper or as powerful as a hurricane.

Why Does Air Move?

The sun heats the Earth's surface, but it does not heat it evenly. Dark surfaces like pavement absorb more heat and get hotter. Light surfaces and water stay cooler. The air above hot spots warms up and becomes lighter, so it rises. Cooler, heavier air nearby rushes in to take the warm air's place. This movement of air is wind. The bigger the temperature difference between two areas, the stronger the wind blows.

Wind in Our Lives

Wind affects our daily lives in many ways. It carries seeds to new places, pushes clouds across the sky, creates waves on the ocean, and can even generate electricity through wind turbines. People have used wind power for thousands of years — from sailing ships to windmills. Understanding wind patterns helps meteorologists predict the weather and warn us about storms.

4 The LEVER Framework

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About LEVER

LEVER is how every ModelIt! lesson works. Students move through five phases to build, test, and refine a computational model. Each lesson is self-contained — students complete all five phases in one session.

L **Locate the System**
Students identify sun heating as an external component and air movement and wind strength as internal components that change based on how much the sun heats the ground.

E **Establish Relationships**
Students determine that more sun heating causes more air movement (positive), and more air movement causes stronger wind (positive).

V **Visualize & Model**
Students build a model showing how the sun's uneven heating creates air movement and wind.

E **Evaluate Outcomes**
Students run hot day and cool day scenarios to compare how heating affects wind strength.

R **Revise & Extend**
Students add surface type or time of day to explore why wind changes throughout the day and in different locations.

Dropping ModelIt! Into Your Existing Unit

ModelIt! is a **supplement**, not a replacement for your science curriculum. Each lesson stands alone, but it becomes even more powerful when students revisit their model throughout a unit. Here are easy ways to weave it in:

At the start of a unit	Use this lesson to introduce the phenomenon. Students build a “first draft” model based on what they already know. Save it.
After a reading or lab	Have students reopen their model and update it: “Based on what you just learned, would you add a component? Change a relationship?” Takes 10–15 minutes.
As a “what-if” tool	When students ask questions during class, say: “Great question — go test that in your model!” Students run new scenarios and report back.
For peer discussion	Partners compare models: “Why does your model have this arrow and mine doesn’t?” Differences spark scientific debate using evidence.
As a presentation tool	Students project their model and walk the class through it live — showing components, running scenarios, answering audience “what-if” challenges in real time.
At the end of a unit	Compare Day 1 model to final model. The growth IS the assessment — students see how their thinking evolved.

5 Slide-by-Slide Facilitation

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Slide 1: Cover

2 min

Say: *"Can you see wind? No! But you can FEEL it. What do you think makes the air move?"*

Watch For

Turn on a small fan. Ask: Is there wind in our classroom now? How do you know?

Slide 2: What Will We Learn Today?

3 min

Say: *"Today we are going to solve the mystery of what makes wind blow!"*

Watch For

Read objectives together. Introduce wind, temperature, and weather with examples.

Slide 3: The Big Question

3 min

Say: *"Wind is invisible — so what is it really? And where does it come from?"*

Watch For

Hold up a tissue near a fan to show wind in action. Ask: What is pushing the tissue?

Slide 4: Let's Build a Model!

3 min

Say: *"We are going to build a model to figure out how the sun creates wind."*

Watch For

Preview LEVER steps. Quick demo: hold hands above a warm surface and a cool surface — feel the difference.

Slide 5: Activity 1: Sort the Parts

7 min

Say: *"What causes wind? The sun heats the ground. The air starts to move. We feel wind! Which part starts it all?"*

Watch For

Guide sorting: Sun Heating is external (it starts the process), Air Movement and Wind Strength are internal.

Slide 6: Activity 2: Connect with Arrows

7 min

Say: *"When the sun heats the ground MORE, does the air move more or less? Does the wind get stronger or weaker?"*

Watch For

Draw positive arrows from Sun Heating to Air Movement and from Air Movement to Wind Strength.

Slide 7: Activity 3: Run the Simulation!

8 min

Say: *"Let us test a hot sunny day and a cool cloudy day. Predict first: which one will be windier?"*

Watch For

Students predict, then run simulations. Compare wind strength in each scenario. Discuss the beach breeze scenario.

Slide 8: What Did We Discover?

7 min

Say: *"Now you know the secret of wind — it is all about the sun heating things up unevenly! Warm air rises, cool air rushes in, and that is WIND!"*

Watch For

Go outside briefly to observe wind clues — flags, tree branches, leaves. Record wind observations.

6 Answer Key & Expected Responses

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Component Sorting

- **EXTERNAL:** Sun Heating
- **INTERNAL:** Air Movement, Wind Strength

Relationships

- **Sun Heating to Air Movement:** POSITIVE (+) — When the sun heats the ground more, the air above it warms up and rises faster. This creates more air movement as cooler air rushes in to replace the rising warm air. More heating means more air circulation.
- **Air Movement to Wind Strength:** POSITIVE (+) — When air moves faster from cool areas to warm areas, the wind is stronger. More air movement means stronger gusts. On calm days when the sun is hidden, there is less air movement and lighter wind.

Key Discoveries

- Wind is moving air — it happens when air flows from one area to another
- The sun heats the ground unevenly — dark areas and land heat up faster than light areas and water
- Warm air rises because it is lighter, and cooler air rushes in to take its place — that rushing air is wind!
- More heating difference means stronger wind — that is why beaches are breezy on hot days

7 Collaborative Learning Protocols

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Why This Matters

ModelIt! is a digital tool, but the deepest learning happens between humans. Computational modeling is most powerful when students build understanding *together* — debating relationships, challenging predictions, and constructing explanations as a community. These protocols ensure every student's voice is part of the model. Use at least ONE per lesson.

Model Gallery Walk

When: After students build models (LEVER Phase V)
Setup: Groups post their models on screens/desks. Half the class stays to "defend," half rotates with sticky notes.
Protocol: Visitors leave 2 sticky notes: one "I notice..." and one "I wonder..." After rotation, defenders read feedback aloud. Class discusses patterns.
Why it works: Students see that the SAME phenomenon can be modeled differently — and that's how real science works. (Bandura: learning through observation of peers)

Scientist Fishbowl

When: During simulation discussion (LEVER Phase E)
Setup: 4-5 students sit in center circle. Rest of class observes with clipboards.
Protocol: Inner circle discusses: "What happened in Scenario 3 and WHY?" using only evidence from their models. Outer circle records strongest arguments. Swap groups.
Why it works: Students practice scientific discourse without teacher directing. Observers learn to evaluate arguments. (Freire: dialogue as the foundation of critical consciousness)

Component Jigsaw

When: During component analysis (LEVER Phase L)
Setup: Assign each group ONE component to become "experts" on. They research it, discuss it, draw it.
Protocol: Regroup so each new team has one expert per component. Experts teach their component to the team. Team builds the model together using everyone's expertise.
Why it works: Every student is essential — the model literally cannot be built without each person's knowledge. Interdependence drives engagement. (Lave & Wenger: learning through legitimate peripheral participation)

Prediction Debate

When: Before running simulations (LEVER Phase E)
Setup: Post a scenario. Students physically move to corners: "Fire Spread will INCREASE a lot" / "INCREASE a little" / "STAY the same" / "DECREASE."
Protocol: Each corner group has 2 minutes to build their argument using model logic. One spokesperson presents. Then RUN the simulation. Losing corners explain what they missed.
Why it works: Movement + social risk + immediate feedback. Students CARE about the outcome because they committed publicly. Wrong predictions become the best learning moments.

Pair Programming for Models (Digital Collaboration)

When: Throughout model building
Setup: Two students, one device. **Driver** controls the mouse/keyboard. **Navigator** directs strategy and catches errors. Switch every 5 minutes.
Protocol: Navigator must verbalize every decision: "I think we should add a negative arrow from Rainfall to Dry Vegetation because..." Driver cannot act without Navigator's reasoning.
Why it works: Forces articulation of scientific thinking. The model becomes a shared artifact, not an individual assignment. Students who struggle with writing can demonstrate understanding verbally. Borrowed from computer science — where the best code is written in pairs.

Quick Protocol Selection Guide

Short on time (5 min)	Prediction Debate — gets bodies moving, generates energy fast
Full period available	Component Jigsaw — deep collaboration, every student contributes

After simulation day	Model Gallery Walk — peer feedback, multiple perspectives
Advanced discussion	Scientist Fishbowl — builds academic discourse skills
Every day default	Pair Programming — transforms individual work into dialogue

8 Differentiation & Extensions

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Below Grade Level

- Pre-sort components as a whole class
- Provide sentence frames for relationships
- Use visual vocabulary cards
- Pair with a strong reader

Advanced Learners

- Add 2-3 additional components
- Research real-world data
- Create multi-step causal chains
- Compare to similar systems globally

English Language Learners

- Pre-teach key vocabulary
- Provide bilingual glossary
- Use visual supports with text
- Allow verbal responses

Gifted & Talented

- Create feedback loop diagrams
- Research prevention strategies
- Write a policy brief
- Connect to climate change

9 Common Misconceptions

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Why This Matters

Addressing misconceptions proactively helps students build accurate scientific understanding. Use these as formative assessment checkpoints during the lesson.

STUDENTS MAY THINK	SCIENTIFIC UNDERSTANDING
"Wind comes from fans or machines"	Natural wind is NOT made by machines — it is created by the sun! The sun heats the Earth unevenly, warm air rises, and cooler air rushes in. That rushing air is wind. Fans copy nature by pushing air with spinning blades, but outdoor wind comes from temperature differences.
STUDENTS MAY THINK	**SCIENTIFIC UNDERSTANDING**
"Wind is only there on stormy days"	Wind happens almost every day, not just during storms! Even on nice sunny days, you can often feel a breeze. Storms have VERY strong wind, but gentle breezes happen whenever there are temperature differences in the air — which is almost always.
STUDENTS MAY THINK	**SCIENTIFIC UNDERSTANDING**
"Cold air makes wind"	Cold air alone does not make wind — the DIFFERENCE between warm and cold air makes wind. You need warm air in one place and cool air in another. The warm air rises and the cool air moves in to replace it. Without a temperature difference, there would be no wind.